| B.E. ARTIFICIAL INTELLIGENCE AND DATA SCIENCE<br>Outcome Based Education (OBE) and Choice Based Credit System (CBCS)<br>**SEMESTER - IV** | | | |
|---|---|---|---|
| **DATA ANALYTICS WITH EXCEL AND R LABORATORY** | | | |
| Course Code | S4CCSA05 | CIE Marks | 50 |
| Teaching Hours/Week (L:T:P) | (0:0:2) | SEE Marks | 50 |
| Credits | 1 | Exam Hours | 1.5 |
| Lecture Hours | - | Practical Hours | 28Hrs |

| **Course objectives:** This course will enable students to: | |
|---|---|
| 1 | **Build** Spreadsheet application by creating structured data |
| 2 | **Automate** the problem solutions using Excel Formulas and advanced features |
| 3 | **Explore** Data Analytics features and Apply analytics using R |

1. Create a spreadsheet for 100 students records which consist of following attributes Student USN, Student Name, course taken , Marks Obtained, Grade points

   a. **Create a suitable examination database and find the sum of the marks(total) of each Student and Average**

   b. Obtain **Letter grades, grade points and corresponding marks range as per below table:**

| Level | Out-standing | Excellent | Very Good | Good | Above Average | Average | Pass | Fail |
|---|---|---|---|---|---|---|---|---|
| Letter Grades | O | A+ | A | B+ | B | C | P | F |
| Grade Points | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 0 |
| Absolute Marks Range (%) | ≥90 | <90 - ≥80 | <80 - ≥70 | <70 - ≥60 | <60 - ≥55 | <55 - ≥50 | <50 - ≥40 | <40 |
| | (90-100) | (80-89) | (70-79) | (60-69) | (55-59) | (50-54) | (40-49) | (0-39) |

**NE:** If a student fails to satisfy attendance (85%) and/or CIE (40%) requirements for the course/s then such course/s shall be marked as NE i.e., Not Eligible to appear for SEE in that course/s.

F' Grade is awarded under the following conditions:

- Failing to secure 35% of marks in SEE (min. 35/100)

   c. **Calculate CGPA and SGPA** as per below rule

The academic performance of a student is indicated by two different indices, Semester Grade Point Average (SGPA) and Cumulative Grade Point Average (CGPA).

- SGPA is an indication of the performance of the student in the current semester.
  SGPA is calculated as indicated below.

$$\textbf{SGPA} = \frac{\Sigma\ [(\text{Course credits}) \times (\text{grade points})] \text{ for all courses with letter grades (in that semester)}}{\Sigma\ [(\text{Course credits})] \text{ for all courses registered in that semester including F grades and excluding W and DP courses.}}$$

- CGPA is an indication of the cumulative performance of the student from the first semester up to the current semester.
  CGPA is calculated as indicated below.

$$\textbf{CGPA} = \frac{\Sigma\ [(\text{Course credits}) \times (\text{grade points})] \text{ for all the courses whose letter grades are E and above from the I semester till the current semester.}}{\Sigma\ [(\text{Course credits})] \text{ whose letter grades are E and above from the I semester till the current semester.}}$$

Calculate SGPA and CGPA for 1st semester, IInd Semester, Supplementary Semester, and for the entire course.

**d. Calculate Percentage equivalence of CGPA. Apply the formula Percentage of Marks = (CGPA - 0.75)x10**

**e. Calculate CGPA range and corresponding class as per below table**

| CGPA | Class |
|---|---|
| ≥5.00 & < 6.75 | Second |
| ≥6.75 & < 7.75 | First |
| ≥7.75 | Distinction |

**Visualization:**

a. Freeze rows and columns
b. Sort the data according to USN.
c. Remove duplicates.
d. Filter: Distinction, First and Second-class data
e. Format the data using different table styles, conditional formatting including colours, labels, or data bars etc.
f. Visualize the data using Charts and Perform result analysis using Grade against Each Subject. Find cases where majority of total value attributed to failure, second class, First class and distinction.
g. Assuming that F grade students are passed in the exam in subsequent semester attempt and modify the data accordingly. Visualize the same over time series of data and do outlier.
h. Apply the following formula suitably and analyse the outcome.
   - IS FUNCTIONS
   - CONDITIONAL FUNCTIONS
   - MATHEMATICAL FUNCTIONS

**Additional – Identify and Learn the short cut key for using above Excel features**

2. Create Sales Database (big data – min 100 records ) to predict the sales of product against quantity sold, Net Value, Gross Value etc. Sample is shown below (100 rows, 32 columns).

| | A | B | C | D | E | F | G | H | I | J | K | L | M | N | P | R | S | T |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | TXN Type | TXN No | TXN Date | Year | Month | Month Name | Day | Day Name | Qtr | FY Qtr | Location Name | Salesman | Salesman Name | Customer | Customer Short Name | Cust Type Name | Cust City | Cust City Nam |
| 2 | INV | 173914 | 02-01-2023 | 2023 | 1 | January | 2 | Wednesday | Qtr 1 | FY Qtr 3 | Warehouse | TNS92 | Ram | HA2161 | Raj | Modern Trade | TMK | Tumakuru |

| Y | Z | AA | AB | AC | AD | AE | AH |
|---|---|---|---|---|---|---|---|
| Item Name | Item Short Name | Item Brand Name | Item Type Name | Item Make | Item Make Name | Item Identification | Grade 1 |
| MQ 100 - Electro Appliances HAND BLENDER - 22111020 | MQ 100 - Electro Appliances HAND BLENDER - | Electro Appliances | Handheld Appliances | GER | Germany | Electro Appliances | DSF19 |

| AK | AL | AM |
|---|---|---|
| Discount | Gross Value | Net Value |
| 0 | 13600 | 13600 |

i. Apply the following formula suitably and analyse the outcome.

- HLOOKUP, VLOOKUP
- FIND, SEARCH
- LEFT, RIGHT, MID, TRIM, CONCANTENATION
- COUNT, COUNTA, COUNTBLANK
- DATE and TIME

ii. Apply Conditional Formatting including labels, colors, data bars etc.

a. Calculate How much Sales for Item # (# can be any specific item in datasheet)
b. Calculate Total Sales in any specific location
c. Calculate Total Sales during week and weekends
d. Apply conditional formatting for quarter wise sales data per salesperson. Note: Sample is shown below

| Salesman Name | Qtr 1 | Qtr 2 | Qtr 3 | Qtr 4 |
|---|---|---|---|---|
| Ram | 2,37,389 | 2,97,355 | 1,91,538 | 1,46,848 |
| Jyothi | 1,96,959 | 4,33,803 | 1,52,971 | 3,64,452 |
| Sree Rav | 85,892 | 1,83,200 | 3,51,871 | 2,48,707 |
| Jaya | 58,688 | 1,11,398 | 17,209 | 1,01,830 |

e. Apply conditional formatting for analysing change in sale. Note: Sample is shown below

| Salesman Name | Qtr 1 | Qtr 2 | 1st Half Sales | Qtr 3 | Qtr 4 | 2nd Half Sales | Change |
|---|---|---|---|---|---|---|---|
| Ram | 2,37,389 | 2,97,355 | 5,34,744 | 1,91,538 | 1,46,848 | 3,38,386 | -37% |
| Jyothi | 1,96,959 | 4,33,803 | 6,30,762 | 1,52,971 | 3,64,452 | 5,17,422 | -18% |
| Sree Rav | 85,892 | 1,83,200 | 2,69,092 | 3,51,871 | 2,48,707 | 6,00,578 | 123% |
| Jaya | 58,688 | 1,11,398 | 1,70,087 | 17,209 | 1,01,830 | 1,19,038 | -30% |

f. Represent the Sales gain (e.g., Red bar for -ve, Blue bar for +ve) through data bar. Note: Sample is given below

| Invoice Date | Sales | Bar |
|---|---|---|
| 12-02-2019 | (12,456) | (12,456) |
| 08-04-2019 | 47,335 | 47,335 |
| 25-09-2019 | 84,133 | 84,133 |

iii. Analyze the data by visualization (apply charts) for Average Price per Month. (Note: Obtain Average price by computing the ratio of net value over quantity)

**Analytics and Visualization :**

a) Format existing data table into pivot table
b) Summarize City and Sales. Sample is shown below.

a.

| Cust City Name | Sum of Quantity | Sum of Net Value |
|---|---|---|
| Bangalore | 15,348 | 23,27,133 |

c) Calculate the City and Sales Contribution. Sample is shown below.

a.

| Cust City Name | Total Qty | Total Sales | WOB |
|---|---|---|---|
| Bangalore | 15,348 | 23,27,133 | 17.29% |

d) Apply the slicers
e) Create multiple pivot table with reference to Customer Type, Salesman, Month wise sales transactions and Customer Account Name(Insert extra column by name Customer Account Name to the dataset). Sample is shown below.

a.

Cust Type Name: E-Commerce | Electro Retail | Modern Tra... | Others | Pharmacy | Traditional ... | Wholesale

| Cust City Name | Total Sales |
|---|---|
| Bangalore | 23,27,133 |

| Salesman Name | Total Sales |
|---|---|
| Akash | 13,96,901 |

| Month | TXN Date | Total Sales |
|---|---|---|
| Jan | | 10,93,218 |

| Cust Account Name | Tota |
|---|---|
| Abu Dhabi Co Operative | |

f) Outline form - Filter, Sort, Group and Apply advanced conditional formatting to the pivot table
g) Create Pivot Report - Analyse and Visualize sales
h) Create Pivot table with external data – show in tabular form

Use Slicers, auto filters etc

3. Create worksheet with following fields: Empno, Ename, Basic Pay(BP), Travelling Allowance(TA), Dearness Allowance(DA), House Rent Allowance(HRA), Income Tax(IT), Provident Fund(PF), Net Pay(NP). Apply appropriate Excel formulas to,

a) Calculate DA , where DA = 50% of basic
Analyze and Visualize, total DA obtained in FY. Apply the conditional formatting for highest DA paid for an employee.

b) Calculate Total HRA received in a given FY

Analyze,

How much HRA can be claimed?

The HRA exemption is calculated as the minimum of the following three amounts:

i) Actual HRA received from the employer

ii) Rent Paid – 10% of Basic Salary

Visualize HRA claims of employee. Apply conditional formatting for minimum and maximum claims.

c) Calculate Income Tax

The tax rate for a salaried individual (under 60 years) as per the old regime is as follows:-

The tax rates for FY 2023-24, as per the new tax regime is, as follows:-

| Tax Slab | Tax Rate |
|---|---|
| Upto Rs 3,00,000 | Nil |
| Rs. 3,00,001-6,00,000 | 5% |
| Rs. 6,00,001-9,00,000 | 10% |
| Rs. 9,00,001-12,00,000 | 15% |
| Rs. 12,00,01-15,00,000 | 20% |
| Above Rs. 15,00,000 | 30% |

Analyze how to save Income tax for a salaried individual?

To reduce taxes, it is highly important to understand the salary structure:-

- Taxable Salary Income = Salary (-) Exemptions (Exempt allowances + Standard deduction +Professional Tax)
- Net taxable income= Taxable Salary Income (-) Deductions under chapter VIA

Visualize the deductions and apply the conditional formatting appropriately

d) Calculate Provident Fund

Assume, Employees' portion of the EPF = 12% * (Basic Salary + DA)

Analyze and Visualize,

i) Employee contribution and Employer Contribution for Basic + DA <= Rs. 12000

ii) Employee contribution and Employer Contribution for Basic + DA >= Rs. 12000

Sample is shown below:

**Employee Provident Fund Calculator (EPF)**

| | | | |
|---|---|---|---|
| Employee Name: | | | |
| Rate of increase of Basic Pay: | 5.00% | Interest Rate | 8.65% |
| Your Contribution: | 12.00% | CTC | 30000 |
| Company's Contribution: | 3.67% | Basic Pay + DA: | 12000 |

| Year | Opening Balance | Basic Pay | Employee Yearly Contribution | Company Yearly Contribution | Total Contribution | Rate of Interest | Closing Balance |
|---|---|---|---|---|---|---|---|
| 1 | 0.00 | 12000.00 | 17280.00 | 5283.76 | 22563.76 | 8.65% | 24515.53 |
| 2 | 24515.53 | 12600.00 | 18144.00 | 5548.00 | 23692.00 | 8.65% | 52377.48 |

e) Calculate Net pay.

Net Salary: Gross Salary – Deductions

Apply conditional formatting for highest net salary and lowest net salary and visualize the data.

4. i) Animate the charts using VBA macro

Consider Sales details for the year 2024 and 2025.

Develop animated Bar chart with data range of Month on X – Axis and Sales on Y –Axis correspondingly. The chart should contain title, label, and legends.

ii) Automated Gantt Chart and Cost Estimation Generator using VBA in Excel.

The objective is to automate the creation of a Gantt chart and total project budget estimation from a task list using a VBA macro. The macro should dynamically create a timeline, highlight task durations in a Gantt-style format, and calculate the overall project cost. Analyze the timeline and cost obtained

**Open Ended Project** – Students are informed to download xlsx or csv files with large dataset (both structured and unstructured data) from Kaggle or GitHub or any online repositories. Import the files and analyze the data using Excel formulas and VBA macros. Use Slicers, Auto Filter, Conditional Formatting and Excel Data Analytics.

**Data Analytics using R**

5 Assess the Financial Statement of an Organization being supplied with 2 vectors of data: Monthly Revenue and Monthly Expenses for the Financial Year. You can create your own sample data vector for this experiment) Calculate the following financial metrics:

a. Profit for each month.

b. Profit after tax for each month (Tax Rate is 30%).

| | |
|---|---|
| | c. Profit margin for each month equals to profit after tax divided by revenue.<br>d. Good Months – where the profit after tax was greater than the mean for the year. e. Bad Months – where the profit after tax was less than the mean for the year.<br>f. The best month – where the profit after tax was max for the year.<br>g. The worst month – where the profit after tax was min for the year.<br>Note: a. All Results need to be presented as vectors<br>b. Results for Dollar values need to be calculated with $0.01 precision, but need to be presented in Units of $1000 (i.e 1k) with no decimal points<br>c. Results for the profit margin ratio need to be presented in units of % with no decimal point.<br>d. It is okay for tax to be negative for any given month (deferred tax asset)<br>e. Generate CSV file for the data |
| 6 | **Employee Productivity**<br>Create a data frame with employee ID, name, department, age, and productivity score.<br>a) Subset employees from the Sales department over 40 with a productivity score above 80.<br>b) Merge with a second data frame containing employee ID and weekly working hours.<br>c) Find correlation between age and productivity (Pearson/Spearman). Plot and log-transform the data. |
| 7 | **Student Performance in Analytics**<br>Create a data frame with student ID, name, course marks, grade, and project score.<br>a) Filter students with project score > 85 and grade 'A'.<br>b) Merge with attendance data using student ID.<br>c) Identify top predictors of course marks and handle multi collinearity. |
| 8 | **Online Course Platform**<br>Create a data frame with course ID, title, enrollment, rating, and instructor.<br>a) Filter courses with rating > 4.5 and enrollments > 1000.<br>b) Merge with course ID and completion rate.<br>c) Visualize relationships, remove correlated variables like rating and completion rate. |
| 9 | **Smart City Traffic Monitoring**<br>Create a data frame with junction ID, vehicle count, average speed, and accident rate.<br>a) Subset records with high traffic and low speed.<br>b) Merge with road type using junction ID.<br>c) Find important features for accident prediction and handle collinearity. |

| **Course Outcomes:**<br>Upon completion of this course the student will be able to: | |
|---|---|
| 1 | Use advanced functions and productivity tools to assist in developing worksheets |
| 2 | Manipulate data lists using Outline and PivotTables. |
| 3 | Use Consolidation to summarize and report results using R programming language. |
| 4 | Identify and Solve big data related problems using Excel Analytics and R |

**Course Articulation Matrix (CO-PO and CO_PSO MAPPING)**

| Course Outcomes | Program Outcomes | | | | | | | | | | | PSOs | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 1 | 2 | 3 |
| CO1 | | 2 | | | | | | | | | | | | 2 |
| CO2 | | 2 | | | | | | | | | | | | 2 |
| CO3 | | 2 | | | | | | | | | | | | 2 |
| CO4 | | 2 | | | | | | | | | | | | 2 |
| **Overall CO** | | **2** | | | | | | | | | | | | **2** |

**Program Articulation Matrix:**

| Course Outcomes | Program Outcomes | | | | | | | | | | | PSOs | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 1 | 2 | 3 |
| | | **2** | | | | | | | | | | | | **2** |